

PULVISIONIDE INHALATION CAPSULE 200 MCG

CONTENT
Each metered dose (inhalation capsule) contains Budesonide 200 mcg

DESCRIPTION
White powder filled in brown clear - colourless capsule no.3
Dry powder inhaler (DPI)

PHARMACODYNAMICS
Pharmacotherapeutic group: Glucocorticoids. ATC code: R03BA02.
Budesonide is a glucocorticosteroid which possesses a high local anti-inflammatory action.

Topical anti-inflammatory effect
The exact mechanism of action of glucocorticosteroids in the treatment of asthma is not fully understood. Anti-inflammatory actions, such as inhibition of inflammatory mediator release and inhibition of cytokine-mediated immune response are probably important.

Onset of effect
After a single dose of orally inhaled budesonide, delivered via dry powder inhaler, improvement of the lung function is achieved within a few hours. After therapeutic use of orally inhaled budesonide delivered via dry powder inhaler, improvement in lung function has been shown to occur within 2 days of initiation of treatment, although maximum benefit may not be achieved for up to 4 weeks.

Airway reactivity
Budesonide has also been shown to decrease airway reactivity to histamine and methacholine in hyper-reactive patients.

Exercise-induced asthma
Therapy with inhaled budesonide has effectively been used for prevention of exercise-induced asthma.

PHARMACOKINETICS
Absorption
After oral administration of budesonide, peak plasma concentration is achieved in about 1-2 hours and the absolute systemic availability is 6-13%. In plasma, 85-95% of budesonide is bound to proteins. In contrast, peak plasma concentration is reached approximately 30 minutes after inhalation. Most of budesonide delivered to the lungs is systemically absorbed.

Distribution
Budesonide is approximately 85-90% bound to plasma protein. The volume of distribution is approximately 3 litres/kg.

Biotransformation and Elimination
Budesonide is mainly eliminated by metabolism. Budesonide is rapidly and extensively metabolised in liver via cytochrome P4503A4 to two metabolites. The in vitro glucocorticoid activity of these metabolites is less than 1% of that of the parent compound. Negligible metabolic inactivation has been observed in human lung and serum preparations. Budesonide is secreted in urine and faeces in the form of conjugated and non-conjugated metabolites.

Linearity
The kinetics of budesonide are dose-proportional at clinically relevant doses.

Paediatric population
Budesonide has a systemic clearance of approximately 0.5L/min in 4-6 years old asthmatic children. Per kg body weight children have a clearance which is approximately 50% greater than in adults. The terminal half-life of budesonide after inhalation is approximately 2.3 hours in asthmatic children. This is about the same as in healthy adults.

Special patient populations
The exposure to budesonide may be increased in patients with liver disease.

INDICATIONS
Indicate for patients with bronchial asthma who require maintenance treatment with glucocorticosteroids for control of the underlying airways inflammation.

RECOMMENDED DOSE
Adults
200 mcg twice daily in the morning and in the evening. During periods of severe asthma, the daily dosage can be increased up to 1600 mcg. In well controlled patients, the daily dose may be reduced below 400 mcg, but should not go below 200 mcg.

Children 6 years and above
200-800 mcg daily, divided into two to four administrations. The highest recommended dose in children is 400 mcg twice daily.

Maintenance
Use the lowest possible dose to control symptoms.
Rinse mouth after each use.

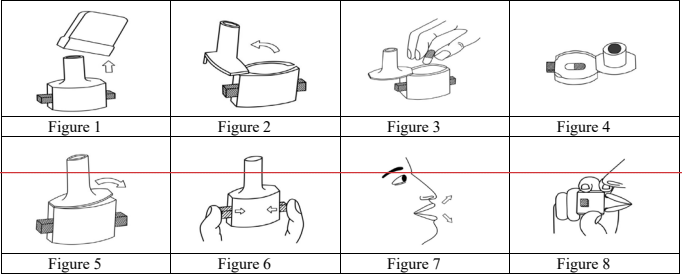
Patients inadequately controlled by bronchodilator therapy:
The use of inhaled budesonide in patients who have never taken steroids or taken only occasional courses of steroids is straightforward. An improvement in respiratory function is normally obvious within a week. The few patients who do not respond during this period usually have excessive mucous in their bronchi so that the drug is unable to penetrate to its site of action. In such cases, a short course of systemic steroid in relatively high dosage should be given to control secretion of mucous and other inflammatory changes in the lungs. Continuation of treatment with inhaled budesonide usually maintains the improvement achieved, the oral steroid being gradually withdrawn. Exacerbation of asthma caused by infection is usually controlled by appropriate antibiotic treatment, by increasing the dose of inhaled budesonide and if necessary, by giving a systemic steroid.

Patients being treated with oral corticosteroids:
The patient should be in a reasonably stable state before being inhaled budesonide in addition to the usual maintenance dose of systemic steroid. After about a week, gradual withdrawal of the systemic steroid is started by reducing the daily dose by one milligram prednisolone or its equivalent, at not less than weekly intervals.

MODE OF ADMINISTRATION
PULVISIONIDE capsules should only be used with the inhaler and inhaled through your mouth (oral inhalation).
Do not intake or swallow PULVISIONIDE capsules.
PULVISIONIDE capsules should always be stored in blister and only removed immediately before use. The capsules should not be stored in the inhaler.
After receiving orally inhaled budesonide, patients should rinse their mouth with water to remove residual drug in the oropharyngeal area and to reduce the development of fungal overgrowth and infection.

Instruction for using PULVISIONIDE
1) Remove the protection cap from the inhaler. (Figure 1)
2) Hold the base of the inhaler firmly.
3) Open the inhaler by turning the mouthpiece in the direction of the arrows. (Figure 2)
4) Take a capsule and place it in the capsule chamber of the inhaler. (Figure 3 and Figure 4)
5) Check whether the capsule is completely inside the capsule chamber. If not, put the capsule in correctly.
6) Close the inhaler by turning the mouthpiece back to the closed position. (Figure 5)
7) Hold the inhaler upright (with the mouthpiece upwards). Hold the push buttons between thumb and index finger. Use both fingers at the same time to press these buttons once. (Figure 6) The capsule is now evenly punctured.
8) Deeply breathe out (not through the inhaler). (Figure 7) After that, hold the inhaler horizontally.
9) Tilt your head back slightly. Insert the mouthpiece of the inhaler completely into your mouth, behind your teeth and on your tongue. Close your lips tight round the mouthpiece. (Figure 8)
10) Breathe in rapidly but steadily, as deeply and forcefully as you can. The inhaler makes a rattling noise if it is operating properly. This is because the capsule spins in the chamber as the powder disperses.
11) Remove the inhaler from your mouth. Hold your breath for about 5-10 seconds so that your lungs can take up the medication. Breathe out gently.
12) Open the inhaler and check whether the capsule is empty, if not repeat from step 8-11.
13) Open the inhaler and remove the empty capsule. Put the protection cap back on the inhaler.

Instruction for cleaning PULVISIONIDE
You should clean the inhaler of PULVISIONIDE once a week.
1) Open the inhaler by turning the mouthpiece as far as possible in the direction of the arrows.
2) Rinse the mouthpiece with running water.
3) Before using the inhaler again, it should be completely dry.



CONTRAINDICATION
Hypersensitivity to budesonide or to any of its excipient – lactose.

WARNINGS AND PRECAUTIONS
Special caution is necessary in patients with active or quiescent pulmonary tuberculosis, and in patients with fungal or viral infections in the airways.

Non-steroid-dependent patients: A therapeutic effect is usually reached within 10 days. In patients with excessive mucus secretion in the bronchi, a short (about 2 weeks) additional oral corticosteroid regimen can be given initially.

Steroid-dependent patients: When transferral from oral steroids to PULVISIONIDE is started, the patient should be in a relatively stable phase. A high dose of PULVISIONIDE is then given in combination with the previously used oral steroid dose for about 10 days.

After that, the oral steroid dose should be gradually reduced (by example 2.5 milligrams prednisolone or the equivalent each month) to the lowest possible level. In many cases, it is possible to completely substitute PULVISIONIDE for the oral steroid.

During transfer from oral therapy to PULVISIONIDE, a generally lower systemic steroid action will be experienced which may result in the appearance of allergic or arthritic symptoms such as rhinitis, eczema and muscle and joint pain. Specific treatment should be initiated for these conditions. During the withdrawal of oral steroids, patients may feel unwell in a non-specific way, even though respiratory function is maintained or improved. Patients should be encouraged to continue with PULVISIONIDE therapy whilst withdrawing the oral steroid, unless there are clinical signs to indicate the contrary. A general insufficient glucocorticosteroid effect should be suspected if, in rare cases, symptoms such as tiredness, headache, nausea and vomiting should occur. In these cases, a temporary increase in the dose of oral glucocorticosteroids is sometimes necessary.

As with other inhalation therapy, paradoxical bronchospasm may occur, with an immediate increase in wheezing after dosing. If this occurs, treatment with inhaled budesonide should be discontinued immediately, the patient assessed and alternative therapy instituted if necessary.

Patients who have previously been dependent on oral steroids may, as a result of prolonged systemic steroid therapy, experience the effects of impaired adrenal function. Recovery may take a considerable amount of time after cessation of oral steroid therapy, hence oral steroid-dependent patients transferred to budesonide may remain at risk from impaired adrenal function for some considerable time. In such circumstances, HPA axis functions should be monitored regularly.

Acute exacerbations of asthma may need an increase in the dose of PULVISIONIDE or additional treatment with a short course of oral corticosteroids and/or an antibiotic, if there is an infection. The patient should be advised to use a short-acting inhaled bronchodilator as rescue medication to relieve acute asthma symptoms.

PULVISIONIDE is not intended for rapid relief of acute episodes of asthma where an inhaled short-acting bronchodilator is required.

If patients find short-acting bronchodilator treatment ineffective or they need more inhalations than usual, medical attention must be sought. In this situation consideration should be given to the need for or an increase in their regular therapy, e.g., higher doses of inhaled budesonide or the addition of a long-acting beta agonist, or for a course of oral glucocorticosteroid.

Patients who have required high dose emergency corticosteroid therapy or prolonged treatment at the highest recommended dose of inhaled corticosteroids, may also be at risk impaired adrenal function. These patients may exhibit signs and symptoms of adrenal insufficiency when exposed to severe stress. Additional systemic corticosteroid treatment should be considered during periods of stress or elective surgery. These patients should be instructed to carry a steroid warning card indicating their needs. Treatment with supplementary systemic steroids or PULVISIONIDE should not be stopped abruptly.

Systemic effects may occur with any inhaled corticosteroids, particularly at high doses prescribed for long periods. These effects are much less likely to occur with inhalation treatment than with oral corticosteroids. Possible systemic effects include Cushing's syndrome, Cushingoid features, adrenal suppression, growth retardation in children and adolescents, decrease in bone mineral density, cataract, glaucoma and more rarely, a range of psychological or behavioural effects including psychomotor hyperactivity, sleep disorders, anxiety, depression or aggression (particularly in children). It is important, therefore, that the

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dose of inhaled corticosteroid is titrated to the lowest dose at which effective control of asthma is maintained.

Reduced liver function affects the elimination of corticosteroids causing lower elimination rate and higher systemic exposure. Be aware of possible systemic side effects.

The plasma clearance following an intravenous dose of budesonide however was similar in cirrhotic patients and in healthy people. After oral ingestion systemic availability of budesonide was increased by compromised liver function due to decreased first pass metabolism. The clinical relevance of this to treatment with PULVISONIDE is unknown as no data exist for inhaled budesonide, but increases in plasma levels and hence an increased risk of systemic adverse effects could be expected.

Co-treatment with CYP3A inhibitors, e.g., itraconazole, ketoconazole, HIV protease inhibitors and cobicistat-containing products is expected to increase the risk of systemic corticosteroid side effects. Therefore, the combination should be avoided unless the benefit outweighs in increased risk, in which case patients should be monitored for systemic corticosteroid side effects. This is of limited clinical importance for short-term (1-2 weeks) treatment with itraconazole or ketoconazole or other potent CYP3A inhibitors, but should be taken into consideration during long-term treatment. A reduction in the dose of budesonide should also be considered.

Oral candidiasis may occur during the therapy with inhaled corticosteroids. This infection may require treatment with appropriate antifungal therapy and in some patients, discontinuation of treatment may be necessary.

Pneumonia in patients with COPD

An increase in the incidence of pneumonia, including pneumonia requiring hospitalisation, has been observed in patients with COPD receiving inhaled corticosteroids. There is some evidence of an increased risk of pneumonia with increasing steroid dose but this has not been demonstrated conclusively across all studies.

There is no conclusive clinical evidence for intra-class differences in the magnitude of the pneumonia risk among inhaled corticosteroid products.

Physicians should remain vigilant for the possible development of pneumonia in patients with COPD as the clinical features of such infections overlap with the symptoms of COPD exacerbations.

Risk factors for pneumonia in patients with COPD include current smoking, older age, low body mass index (BMI) and severe COPD.

Visual disturbance

Visual disturbance may be reported with systemic and topical corticosteroid use. If a patient presents with symptoms such as blurred vision or other visual disturbances, the patient should be considered for referral to an ophthalmologist for evaluation of possible causes which may include cataract, glaucoma or rare diseases such as central serous chorioretinopathy (CSCR) which have been reported after use of systemic and topical corticosteroids.

Paediatric population

Influence on growth

It is recommended that the height of children receiving prolonged treatment with inhaled corticosteroids is regularly monitored. If growth is closed, therapy should be re-evaluated with the aim of reducing the dose of inhaled corticosteroid, if possible, to the lowest dose at which effective control of asthma is maintained. The benefit of the corticosteroid therapy and the possible risk of growth suppression must be carefully weighed. In addition, consideration should be given to referring the patient to a paediatric respiratory specialist.

INTERACTIONS WITH OTHER MEDICAMENTS

The metabolism of budesonide is primarily mediated by CYP3A4. Co-treatment with CYP3A inhibitors, including cobicistat-containing products, is expected to increase the risk of systemic side effects. The combination should be avoided unless the benefit outweighs the increased risk of systemic corticosteroid side effects, in which patients should be monitored for systemic corticosteroid side effects. If PULVISONIDE is co-administered with anti-fungal (such as itraconazole and ketoconazole), the period between treatments should be as long as possible. A reduction of the budesonide dose could be considered.

Limited data about this interaction for high-dose inhaled budesonide indicate that marked increases in plasma levels (on average four-fold) may occur if itraconazole, 200 mg once daily, is administered concomitantly with inhaled budesonide (single dose of 1000 µg).

Raised plasma concentrations of and enhanced effects of corticosteroids have been observed in women also treated with oestrogens and contraceptive steroids, but no effect has been observed with budesonide and concomitant intake of low dose combination oral contraceptives.

Because adrenal function may be suppressed, an ACTH stimulation test for diagnosing pituitary insufficiency might show false results (low values).

PREGNANCY AND LACTATION

Pregnancy

Most results from prospective epidemiological studies and world-wide post marketing data have not been able to detect an increased risk for adverse effects for the foetus and new-born child from the use of inhaled budesonide during pregnancy. In animal studies, glucocorticosteroids have been shown to induce malformations. This is not likely to be relevant for humans given recommended doses, but therapy with inhaled budesonide should be regularly reviewed and maintained at the lowest effective dose. It is important for both foetus and mother to maintain an adequate asthma treatment during pregnancy. As with other drugs administered during pregnancy, the benefit of the administration of budesonide for the mother should be weighed against the risks to the foetus.

Inhaled glucocorticosteroids should be considered in preference to oral glucocorticosteroids because of the lower systemic effects at the doses required to achieve similar pulmonary responses.

Breast-feeding

Budesonide is excreted in breast milk. However, at the therapeutic doses of PULVISONIDE no effects on the suckling child are anticipated. PULVISONIDE can be used during breast feeding. Maintenance treatment with inhaled budesonide (200 or 400 micrograms twice daily) in asthmatic nursing women results in negligible systemic exposure to budesonide in breast-fed feeding.

UNDESIRABLE EFFECTS

Table 1: Undesirable Drug Effects by System Organ Class (SOC) and Frequency

System Organ Class	Frequency	Undesirable drug effect
Infectious and infestations	Common	Oropharyngeal candidiasis Pneumonia (in COPD patients)
Immune system disorders	Rare	Immediate and delayed hypersensitivity reactions including rash, contact dermatitis, urticaria, angioedema and anaphylactic reaction
Endocrine disorders	Rare	Signs and symptoms of systemic corticosteroid effects, including adrenal suppression and growth retardation*
Psychiatric disorders	Uncommon	Anxiety Depression

	Rare	Psychomotor hyperactivity Sleep disorders Aggression Behavioural changes (predominantly in children) Tremor**
Nervous System Disorders	Uncommon	
Eye disorders	Uncommon	Cataract Vision, blurred
	Not Known	Glaucoma
Respiratory, thoracic and mediastinal disorders	Common	Cough Hoarseness Throat irritation
	Rare	Bronchospasm Dysphonia Hoarseness***
Skin and subcutaneous tissue disorders	Rare	Bruising
Musculoskeletal and connective tissue disorders	Uncommon	Muscle spasm

* refer to Paediatric population, below

** based on the frequency reported in clinical trials

*** rare in children

Occasionally, signs or symptoms of systemic glucocorticosteroid-side effects may occur with inhaled glucocorticosteroids, probably depending on dose, exposure time, concomitant and previous corticosteroid exposure, and individual sensitivity.

Description of selected adverse reactions

The candida infection in the oropharynx is due to drug deposition. Advising the patient to rinse the mouth out with water after each dosing will minimise the risk.

As with other inhalation therapy, paradoxical bronchospasm may occur in very rare cases.

In placebo-controlled studies, cataract was also uncommonly reported in the placebo group.

Clinical trials with 13119 patients on inhaled budesonide and 7278 patients on placebo have been pooled. The frequency of anxiety was 0.52% on inhaled budesonide and 0.63% on placebo; that of depression was 0.67% on inhaled budesonide and 1.15% on placebo.

Paediatric population

Due to the risk of growth retardation in the paediatric population, growth should be monitored.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

PULVISONIDE has no or negligible influence on the ability to drive and use machines.

OVERDOSE AND TREATMENT

Symptoms of overdose

The acute toxicity of budesonide is low. Chronic use in excessive doses can result in systemic glucocorticosteroid effects, such as increased susceptibility to infection, hypercorticism and adrenal suppression. Atrophy of the adrenal cortex can occur and the ability to adapt to stress can be impaired.

Therapeutic management of overdose

For acute overdosage, even in excessive doses, is not expected to be a clinical problem. The treatment with inhaled budesonide should be continued at the recommended dose to control asthma. HPA axis function recovers in a few days.

In stress situations, it may be necessary to administer corticosteroids as a precaution (e.g., high doses of hydrocortisone). Patients with adrenocortical atrophy are regarded as being steroid-dependent and must be adjusted to the adequate maintenance therapy of a systemic corticosteroid until the condition has stabilised.

STORAGE CONDITION

Do not store above 30°C

Replace the mouthpiece cap firmly.

Keep out of the reach of children.

Jauhi daripada kanak-kanak.

DOSAGE FORMS AND PACKAGING AVAILABLE

Clear/colourless PVC/PVdC laminated tropicalised blister covered with peelable aluminium lidding foil of 5 capsules packed in paper carton with/without inhaler.

MANUFACTURER

T.O. CHEMICALS (1979) Ltd.

53-Lieb-Klong 7-Road, Moo 4,

Bueng Kham Phroi, Lam Luk Ka,

Pathumthani 12150, Thailand

PRODUCT REGISTRATION HOLDER

IMEKS Pharma Sdn. Bhd.

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51200 Kuala Lumpur

DATE OF REVISION

7 December 2022

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